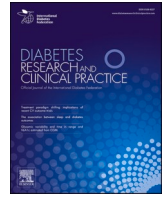




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Prediction of metabolic syndrome: A machine learning approach to help primary prevention

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ABSTRACT

Aims: To describe the performance of machine learning (ML) applied to predict future metabolic syndrome (MS), and to estimate lifestyle changes effects in MS predictions.

Methods: We analyzed data from 17,182 adults attending a checkup program sequentially (37,999 visit pairs) over 17 years. Variables on sociodemographic attributes, clinical, laboratory, and lifestyle characteristics were used to develop ML models to predict MS [logistic regression, linear discriminant analysis, k-nearest neighbors, decision trees, Light Gradient Boosting Machine (LGBM), Extreme Gradient Boosting]. We have tested the effects of changes in lifestyle in MS prediction at individual levels.

Results: All models showed adequate calibration and good discrimination, but the LGBM showed better performance (Sensitivity = 87.8 %, Specificity = 70.2 %, AUC-ROC = 0.86). Causal inference analysis showed that increasing physical activity level and reducing BMI by at least 2 % had an effect of reducing the predicted probability of MS by 3.8 % (95 % CI = -4.8 %; -2.7 %).

Conclusion: ML models based on data from a checkup program showed good performance to predict MS and allowed testing for effects of lifestyle changes in this prediction. External validation is recommended to verify models' ability to identify at-risk individuals, and potentially increase their engagement in preventive measures.

1. Introduction

Metabolic syndrome (MS) is characterized by metabolic changes that increase the risk for diabetes mellitus, cardiovascular disease, neoplasia and cause higher mortality [1–4]. According to the Global Burden of Disease study, the population exposed to metabolic risk factors worldwide was either stagnant or increased between 2010 and 2019 [5]. In US adults, the MS prevalence increased from 25 % in 1988–1994 to 34 % in 2007–2012 [6]. In Brazilian adults, a recent systematic review and meta-analysis estimated a MS prevalence of 26 % in males and 38 % in females in the last decade [7]. Together with the increasing prevalence of obesity and diabetes mellitus, MS is a strong contributor to cardiovascular disease, the main cause of death worldwide [8–11].

Machine learning (ML), an application of artificial intelligence and computer science, uses algorithms that learn and improve accuracy from experience, looking for patterns in data so it can later make inferences

based on the examples provided. In healthcare, it has been applied to predict multiple outcomes, to recognize patterns in images, and even to identify health conditions by interpreting text from clinical notes recorded in electronic health records. ML models have been applied to predict health outcomes using real world data because they are flexible to deal with large volumes of information and multiple variables, considering the potential interactions between them. Models usually learn by training in a dataset in which the outcome is known and, thereafter, the trained model is applied to new data to test its performance [12]. Recent studies have reported the use of ML to predict MS [13–26] based on clinical information, laboratory tests, and genetics. However, most models were developed to predict the current MS risk based on cross-sectional data [14,16–19,21,22,24,26], and evidence on ML algorithms to predict future MS in individuals without the condition is more limited [13,20,23,25]. Also, there is scarce data on how ML can be leveraged to predict the effect of lifestyle modifications on the risk of

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future MS [23].

Given these gaps in the literature, the aims of our study were to develop ML models to predict future MS in adults attending a periodic health examination program and to estimate the effect of reducing body mass index (BMI) and/or increasing leisure-time physical activity in the predicted probability of future MS.

2. Methods

2.1. Study design and population

This observational study was approved by our Local Ethics Review Board. All data were analyzed after a de-identification procedure to ensure data privacy in compliance with guidelines for best practice and current local regulatory legislation. We included data from a cohort of adults who attended a periodic health examination (checkup) in a private health institution in São Paulo, Brazil, from October 2004 to May 2021. This health promotion and prevention program focuses on the screening of health conditions and risk factors, in addition to the identification of behaviors that could be targeted for intervention. In a checkup visit, a set of multidisciplinary assessments (medical history, family history, physical examination, and risk assessment of specific conditions) and diagnostic tests (laboratory, imaging, and cardiac stress test) are performed according to the patient's age and sex. Repeated assessments usually occur every one or two years. Interventions are not implemented in the program, but the checkup results and recommendations are available to be discussed with the patient's physician.

This study used two sequential checkup visits (a pair of visits) as the unit of analysis to take advantage of repeated MS assessments in the cohort. We named "index visit" the pair's first visit, whose data were used as input to predict MS in the subsequent visit. For individuals with more than two sequential visits, multiple pairs of visits were included in the model (the subsequent visit of one pair could be the index visit of another pair). A total of 63,072 individuals had a checkup visit in the study period, summing 127,453 visits. We included in this analysis only pairs of visits from individuals who had sequential visits within 8 to 24 months apart and did not have MS in the index visits, comprising 37,999 pairs of visits from 17,182 individuals (see [Supplementary Fig. S1A and S1B](#)).

2.2. Outcome

The outcome used in the models was the presence of MS in the subsequent visit. MS definition was based on the definition proposed in 2009 by the International Diabetes Federation and the American Heart Association/National Heart, Lung, and Blood Institute [27]. The criteria required the presence of 3 out of the 5 following risk factors: increased waist circumference (≥ 80 cm for women and ≥ 94 cm for men) [28], elevated triglycerides (≥ 150 mg/dL), reduced HDL-cholesterol (< 50 mg/dL for women and < 40 mg/dL for men), elevated blood pressure (systolic blood pressure ≥ 130 mmHg and/or diastolic blood pressure ≥ 85 mmHg) or in use of any antihypertensive drug treatment, increased fasting blood glucose (≥ 100 mg/dL) or in use of any antihyperglycemic treatment.

2.3. Predictors

As we intended to develop models that could be applied in multiple settings, we chose to include parameters frequently assessed in clinical settings that do not require specialized equipment, using a priori knowledge. The following set of variables were extracted from the electronic health record: age, sex, systolic and diastolic blood pressure (in mmHg), waist circumference (in cm), weight (in kg), height (in meters), BMI (calculated as $\text{weight}/\text{height}^2$), medications (lipid-lowering drugs, antidiabetics and antihypertensives), smoking status (current smoker, previous smoker or never smoked), and laboratory

tests results (hemoglobin, hematocrit, leukocytes, platelets, total cholesterol, HDL-cholesterol, LDL-cholesterol, triglycerides, fasting glucose, serum creatinine, serum urea transaminases, gamma-glutamyl transferase, and thyrotropin). Additionally, we used data from self-responded questionnaires applied during the checkup visit containing the following: leisure-time physical activity by the International Physical Activity Questionnaire (classifying the sample in inactive, low activity levels, moderate activity levels, and high activity levels) [29], risky or hazardous consumption of alcohol or any alcohol use disorder by the Alcohol Use Disorder Identification Test [30] and depressive symptoms by the Beck's Depression Inventory [31].

2.4. Basics statistics

Statistical analysis was conducted with Python 3.8.0 (Python Software Foundation, Fredericksburg, Virginia, USA). We used the median (interquartile range) and absolute and relative frequencies to describe numeric and categorical variables, respectively. Characteristics of participants on the index visits were compared using the Wilcoxon rank sum test for numeric variables and the chi-square test for categorical variables. The alpha was set at 0.05.

2.5. Data preparation for machine learning

Numeric variables were checked for unrealistic values and 98 such entries were removed. Median imputation was used to replace missing values. These numeric variables were then normalized using the z-score (calculated by subtracting the sample mean from each individual value and then dividing the difference by the sample standard deviation). Dummy variables were created from the categorical ones by converting as many categories as possible into "yes/no", including dummy variables for missing values. Data was split into three different sets: 70 % for training, 15 % for validation, and 15 % for testing. Individuals did not overlap between these data sets.

2.6. Machine learning models

All the variables described previously were included as predictors to build seven ML models: (1) logistic regression, (2) linear discriminant analysis, (3) k-nearest neighbors, (4) decision tree, (5) random forests, (5) extra trees, (6) Light Gradient Boosting Machine, and (7) Extreme Gradient Boosting. A brief explanation of each ML technique can be seen in the Supplementary Box S1. The models were built using the following libraries available for Python 3.8.0: sklearn 0.23.2 [32], LGBM 3.1.0 [33], and xgboost 1.3.3 [34]. To avoid overfitting, 5-fold cross-validation was performed in the training dataset. The validation dataset was used to select the optimum combination of hyperparameters (parameters used to control the learning process of the models, that are not related to the predictors) for each ML technique (see Supplementary Table S1).

We assessed the predictive performance of each ML model by computing metrics of calibration and discrimination in the test set. Calibration was assessed using the Brier score (ranging from 0 to 1 and lower scores indicate better calibration) [35], and discrimination was measured by the area under the receiver operating characteristic curve (ROC-AUC) and the area under the precision-recall (PR-AUC) curve, whose values range from 0 to 1 and higher values indicate better discrimination. The precision-recall curve summarizes the trade-off between the positive predictive value (also named precision) and the sensitivity (also named recall) for different predicted probability thresholds [36]. We selected the best performing model based on the combination of these three metrics in the following order of importance: the lowest Brier score, the highest ROC-AUC, and the highest PR-AUC. Additionally, we assessed sensitivity, specificity, positive predictive value, and negative predictive value for the best performing model in the test set, considering the predicted probability threshold that

maximizes both sensitivity and specificity (Youden Index). Finally, we used SHAP (SHapley Additive exPlanations) to explore the relevant predictors of the best performing model. SHAP is used to understand how complex models use the data to produce predictions. It considers the interactions between the predictors and how they affect the outcome, providing, for each predictor, its order of importance, the strength, and the direction of the association, here represented in the SHAP summary plot [37].

2.7. Estimation of the effects of lifestyle modifications

The conditional average treatment effect (CATE) is a causal inference approach to estimate the effect of an intervention using observational data, assuming confounding by other variables [38]. We assessed the effect of the following interventions: a) BMI reduction by at least 2 %, b) increment in at least one level of physical activity measured by the International Physical Activity Questionnaire, and c) both interventions. In a population of individuals with similar predictors, the CATE measures the difference in the probability of future MS between those who had each of the interventions mentioned above and those who had not [39]. Such probabilities were estimated for each intervention by the best performing ML model, and the CATE was calculated in the test set using the S-learner technique implemented by the CausalML library [40].

2.8. Secondary analyses

We also trained and tested the best performing model in two additional analyses to assess its performance across different groups: (1) to predict future MS separately among men and women, because MS risk factors differ according to sex [41]; (2) to predict MS in the subsequent visit in a larger sample of individuals that had attended the periodic health examination program, regardless of the MS status in the index visit (we included in this analysis 46,336 pairs of visits from 19,561 individuals).

3. Results

3.1. Prediction of future metabolic syndrome

The mean age of the sample of index visits was 51.2 ± 9.7 years and 25.5 % were female. MS was diagnosed in 3,196 subsequent visits (8.4 %). Alterations in one or two parameters that define MS were present in 13,618 (35.8 %) and 9,885 (26.0 %) index visits, respectively. Table 1 shows the characteristics of the total sample and according to MS status in the subsequent visit. Individuals who developed MS in the subsequent visit were older, predominantly male, less physically active, more frequently previous smokers, had higher BMI, and had higher levels of total-cholesterol, LDL-cholesterol, transaminases, and gamma glutamyl transferase.

The performance to predict future MS of each ML model technique in the test set (measured by the calibration and discrimination) can be seen in Table 2. The model that achieved the best performance was the Light Gradient Boosting Machine and its calibration and discrimination results (Brier score = 0.065; ROC-AUC = 0.858; PR-AUC = 0.336) can be seen in Fig. 1. Its sensitivity, specificity, positive predictive value, and negative predictive value were 87.8 %, 70.2 %, 21.3 %, and 98.4 %, respectively. The SHAP summary plot for the Light Gradient Boosting Machine depicts the most relevant predictors and their impact on the outcome (Fig. 2 and Supplementary Fig. S2). The MS parameters were included among the 10 most relevant predictors. Other relevant features were BMI, age, male gender, and gamma glutamyl transferase values.

3.2. Estimation of the effects of lifestyle modifications

The histogram depicting the CATE estimate for the three different interventions is shown in Fig. 3. Note that a negative CATE is desirable

Table 1

Characteristics of the total sample of index visits and according to metabolic syndrome status in the subsequent visit.

	Total sample (n = 37,999)	Without metabolic syndrome (n = 34,803)	With metabolic syndrome (n = 3,196)
Age (years), median (IQR)	51 (44–58)	50 (44–57)	55 (48–62)
Female, n (%)	9709 (25.6)	9316 (26.8)	393 (12.3)
Waist circumference (cm), median (IQR)*	92 (87–99)79	92 (87–98)78	99 (93–104)89
Men	(73–85)	(73–85)	(82.5–95.8)
Women			
Systolic blood pressure (mmHg), median (IQR)*	114 (110–120)	112 (110–120)	120 (113–125.5)
Diastolic blood pressure (mmHg), median (IQR)*	76 (70–80)	75 (70–80)	80 (75–80)
Anti-hypertensive medication, n (%)*	3233 (9.3)	2551 (8.0)	682 (24.0)
HDL-cholesterol (mg/dL), median (IQR)*	60 (51–70)47	60 (52–71)48	51 (44–56)43
Women	(42–55)	(42–55)	(38–48)
Men			
Triglycerides (mg/dL), median (IQR)*	97 (73–130)	94 (71–126)	131 (105–166)
Fasting blood glucose (mg/dL), median (IQR)*	85 (80–91)	85 (80–90)	89 (84–95)
Anti-diabetic medication, n (%)*	485 (1.6)	378 (1.4)	107 (4.3)
Body mass index (kg/m ²), median (IQR)*	25.4 (23.4–27.5)	25.2 (23.3–27.2)	27.6 (25.9–29.9)
Smoking status, n (%)*	2621	2377	244
Current smoker	(7.5)4838	(7.4)4258	(8.6)580
Previous smoker	(13.9)	(13.3)	(20.4)
Leisure-time physical activity, n (%)*	5366	4790	576
Inactive	(15.5)	(15.1)9120	(20.6)951
Low activity levels	10,071	(28.7)	(34.0)1045
Moderate activity levels	(29.2)	13,413 (42.3)	(37.4)226
High activity levels	14,458	4423	(8.1)
AUDIT, median (IQR)*	(41.9)4649	(13.9)	
Beck's Depression Inventory, median (IQR)*	(13.5)		
AUDIT, median (IQR)*	3 (2–5)	3 (2–5)	3 (2–5)
Beck's Depression Inventory, median (IQR)*	4 (1–7)	4 (1–7)	4 (1–8)
Total cholesterol (mg/dL), median (IQR)*	188 (167–212)	188 (166–212)	192 (170–218)
LDL-cholesterol (mg/dL), median (IQR)*	115 (95–137)	115 (94–137)	121 (100–144)
Lipid-lowering medication, n (%)*	4779 (13.8)	4135 (13.0)	644 (22.6)
Creatinine (mg/dL), median (IQR)*	0.9 (0.79–1.0)	0.9 (0.79–1.0)	0.9 (0.8–1.0)
Urea (mg/dL), median (IQR)*	34 (29–39)	34 (29–39)	35 (30–40)
AST (U/L), median (IQR)*	26 (22–31)	26 (22–31)	28 (24–33)
ALT (U/L), median (IQR)*	34 (26–44)	33 (26–43)	40 (32–51)
GGT (U/L), median (IQR)*	25 (18–36)	24 (18–35)	31 (24–44)
Hemoglobin (g/dL), median (IQR)*	14.6 (13.7–15.3)	14.5 (13.6–15.3)	15.0 (14.2–15.6)
Hematocrit (%), median (IQR)*	42.7 (40.4–44.7)	42.6 (40.2–44.6)	43.7 (41.7–45.7)
Leucocytes (*10 ³ /μL), median (IQR)*	6.02 (5.20–7.01)	6.0 (5.2–7.0)	6.37 (5.50–7.36)
Platelets (*10 ³ /μL), median (IQR)*	225 (195–258)	225 (195–258)	225 (194–259)
Thyrotropin (μIU/mL), median (IQR)*	2.00 (1.40–2.81)	2.00 (1.40–2.80)	2.05 (1.45–2.91)

ALT: alanine aminotransferase; AST: Aspartate aminotransferase; AUDIT = Alcohol Use Disorders Identification Test; GGT: gamma-glutamyl transferase;

HDL: high-density lipoprotein; IQR: interquartile range; LDL: low-density lipoprotein.

* There were 3765 missing values for waist circumference, 3816 for systolic blood pressure, 3872 for diastolic blood pressure, 3313 for anti-hypertensive medication, 4042 for HDL-cholesterol, 4033 for triglycerides, 3791 for fasting blood glucose, 3812 for body mass index, 3215 for smoking status, 3455 for leisure-time physical activity, 7788 for anti-diabetic medication, 3308 for lipid-lowering medication, 4273 for AUDIT score, 4554 for Beck's depression inventory, 4031 for total cholesterol, 4033 for LDL-cholesterol, 8518 for creatinine, 8250 for urea, 4043 for AST, 4051 for ALT, 4094 for GGT, 4043 for hemoglobin, 4030 for hematocrit, 4049 for leucocytes, 4382 for platelets, and 8545 for thyrotropin.

Table 2

Calibration and discrimination of each machine learning model technique in the test set.

Classifier	Brier score	ROC-AUC	PR-AUC
Logistic Regression	0.068	0.840	0.280
LDA	0.072	0.834	0.263
K Nearest Neighbors	0.069	0.808	0.248
Decision Tree	0.069	0.802	0.254
Random Forest	0.065	0.850	0.324
Extra Trees	0.067	0.843	0.303
LGBM	0.065	0.858	0.336
XGBoost	0.065	0.857	0.328

LDA: linear discriminant analysis; LGBM: light gradient boosting machine; PR-AUC: area under the precision-recall curve; ROC-AUC: area under the receiver operator characteristic curve; XGBoost: extreme gradient boosting.

in this case, as it shows a decrease in the probability of future MS. Individuals who had a decrease in BMI of at least 2 % had an absolute reduction in the probability of future MS in the subsequent visit of 3.7 % (95 % CI −4.6 %; −2.9 %) when compared to individuals who had not had this intervention. Individuals with an increase in at least one level in the physical activity scale had an absolute reduction in the predicted probability of future MS of 0.2 % (95 % CI = −1.3 %; 0.8 %). Finally, both decreasing BMI by at least 2 % and increasing at least one level in the physical activity scale reduced the probability of future MS at individual level by 3.8 % (95 % CI = −4.8; −2.7 %).

3.3. Prediction of future metabolic syndrome stratified by sex

Overall, the Light Gradient Boosting Machine showed a slightly poorer performance to predict future MS separately in women (Brier score = 0.038, ROC-AUC = 0.852, PR-AUC = 0.181) and men (Brier

score = 0.082, ROC-AUC = 0.821, PR-AUC = 0.316) than the model that was trained in the whole sample (Supplementary Fig. S3). The sensitivity, specificity, positive predictive value, and negative predictive value were 78.1 %, 79.9 %, 14.8 %, and 98.8 %, respectively for women and 86.8 %, 65 %, 22.5 %, and 97.7 %, respectively for men. The 10 most relevant predictors were similar for both sexes and included the MS parameters, age, leisure-time physical activity, and BMI (Supplementary Figs. S4 to S7).

3.4. Prediction of metabolic syndrome regardless of metabolic status in the index visit

When we assessed the prediction of MS in a subsequent checkup visit for individuals with and without MS at the index visit, the Light Gradient Boosting Machine presented better discrimination (ROC-AUC = 0.896; PR-AUC = 0.641) and it was slightly less calibrated (Brier score = 0.089) than the best performing predictive model of future MS (Supplementary Fig. S8). The sensitivity, specificity, positive predictive value, and negative predictive value were 82.5 %, 80.6 %, 46.4 %, and 95.8 %, respectively. The 10 most relevant predictors were MS parameters, antihypertensive use, BMI, age, and anti-diabetic medication (Supplementary Fig. S9). Finally, the histograms of the distribution of the CATE estimates for the three different interventions can be seen in the Supplementary Fig. S10. The effects of a decrease in BMI by at least 2 %, an increase in the physical activity level, and of both interventions were an absolute reduction in the predicted probability of future MS of 3.4 % (95 % CI = −4.5 %; −2.3 %), 0.3 % (95 % CI = −1.6 %; 0.9 %), and 3.1 % (95 % CI = −4.8 %; −1.4 %), respectively.

4. Discussion

In this study, we developed and tested ML models to predict future MS in adults who attended sequential periodic health examinations. Although most developed algorithms presented good predictive performance, the Light Gradient Boosting Machine achieved the best combination of calibration and discrimination metrics. We were also able to apply ML to estimate the effect size of lifestyle modifications (BMI reduction and/or improvement in physical activity levels) in the predicted probability of future MS in the individual level, by calculating the CATE, a metric derived from causal inference techniques.

Considering its close association with diabetes mellitus and cardiovascular disease, MS is a public health issue. Importantly, studies have demonstrated the efficacy of lifestyle interventions to reverse MS [42–44]. A challenging task is to increase awareness of asymptomatic

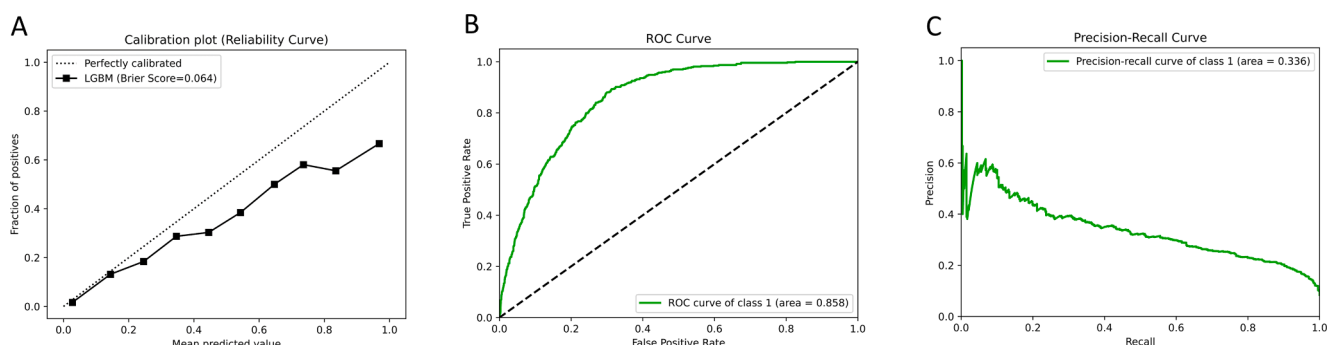


Fig. 1. Calibration and discrimination of the Light Gradient Boosting Machine to predict future metabolic syndrome. **A:** Calibration plot depicting the difference between the probability of future metabolic syndrome predicted by the model and the real frequency of future metabolic syndrome (fraction of positives) in the study sample. The dotted line represents the perfect calibration of the model. The Brier score ranges from 0 to 1, with lower values indicating better calibration. **B:** Receiver operating characteristic curve, depicting the tradeoff between sensitivity and the false positive rate of the model according to the range of predicted probabilities of future MS. The area under the curve represents the model skill to discriminate between the occurrence or not of future MS. A value of 0.5 is used as reference for a random estimator (represented by the dotted line), while a perfect model would present a value of 1.0. **C:** Precision-recall curve, depicting the tradeoff between precision (also known as positive predictive value) and recall (also known as sensitivity) for different predicted probability thresholds. A high area under the curve represents both high recall (low false positive rate) and high precision (low false negative rate).

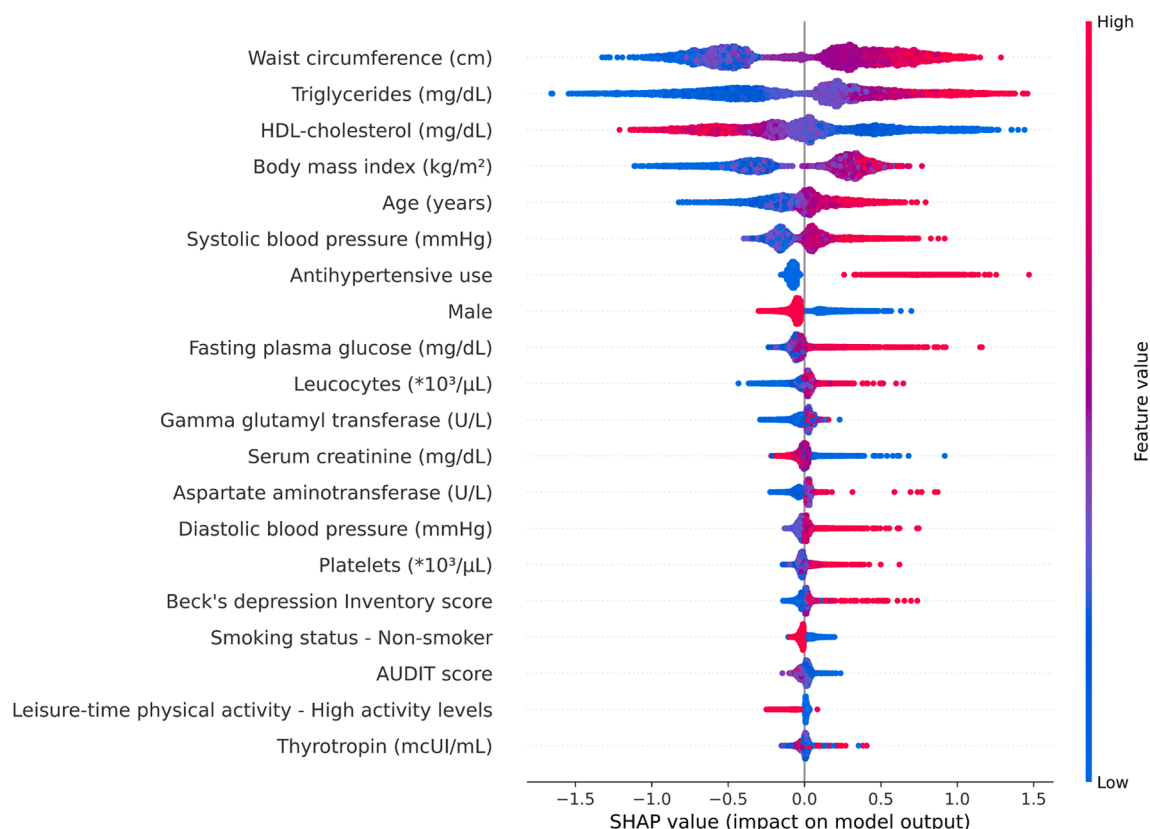


Fig. 2. SHapley Additive exPlanations (SHAP) summary plot for the Light Gradient Boosting Machine developed to predict future metabolic syndrome: graphic representation of how predictors impact the outcome. The lines correspond to the predictors, whose order of importance on the prediction of the outcome decreases from top to bottom. The dots that compose the lines represent the SHAP value for the predictors in each index checkup visit. The combination of the color and the position of the dots in the line indicates the strength and direction of the association: (1) red indicates a higher value and blue indicates a lower value of the predictor, respectively. In the case of categorical variables, red indicates the presence of a specific category of the predictor and blue indicates the absence of this specific category; (2) dots on the left side of the chart represent a decreased risk of the outcome (the more to the left the dot is, the higher the impact on decreasing the outcome risk) and dots on the right side represent an increased risk of the outcome (the more to the right the dot is, the higher the impact on increasing the outcome risk).

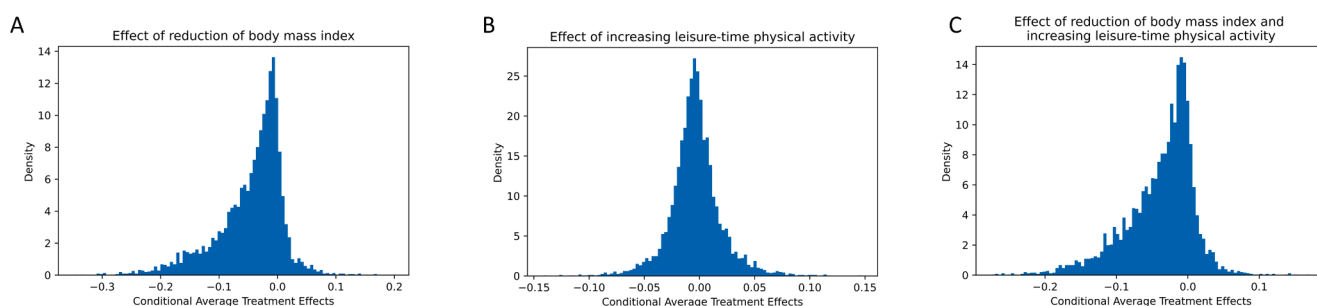


Fig. 3. Histogram of the distribution of conditional average treatment effects (CATE) of lifestyle modifications on the prediction of future metabolic syndrome. A: Effect of reducing body mass index; B: Effect of increasing physical activity levels measured by the International Physical Activity Questionnaire; C: Effect of reducing body mass index and increasing physical activity levels.

metabolic abnormalities and engagement in lifestyle modifications [45]. Moreover, as previously demonstrated in our population, individuals frequently perceive their cardiovascular risk as lower than the actual calculated risk, which could also happen with the risk of MS [46]. In this context, the clinical application of a ML tool trained with real world data that provides a personalized prediction of future MS and simulates the effect of lifestyle modifications in such risk prediction, might increase the chance of motivating at-risk individuals to adopt healthier behaviors and, ultimately, prevent bad outcomes.

The performance in predicting future MS by ML models in our work is in agreement with similar reports in the literature. Karimi-Alavijeh et

al developed fairly accurate models (decision tree and support vector machine) to predict future MS in adults without cardiovascular conditions after 7 years of follow-up [20]. A model based in an artificial neural network presented higher sensitivity (95 % vs 27 %) and similar specificity (93 % vs 91 %) when compared to logistic regression to predict future MS after 6 years of follow-up in men participating in annual periodic health examinations [13]. However, these articles did not report other relevant metrics to assess the performance of the models, such as discrimination and calibration [13,20]. In a retrospective cohort of periodic health examinations, logistic regression, artificial neural networks, classification and regression trees, and support vector machine

showed fair discrimination (AUC-ROC between 0.74 and 0.78) to predict future MS after 4 years of follow-up. Models were based on the MS components, age, BMI, and additional laboratory tests [25]. Finally, a Extreme Gradient Boosting model also had good discrimination (AUC-ROC = 0.879) to predict MS in two years in a prospective population-based cohort of adults [23].

As expected, we found that the combination of decreasing the BMI and increasing physical activity was the most impactful intervention to reduce the probability of future MS. On the other hand, the average effect of only incrementing physical activity, without weight reduction, was nearly zero. We hypothesize that this happened because the International Physical Activity Questionnaire asks specifically about the levels of physical activity during the week preceding the index visit, precluding a reliable assessment of physical activity during weeks or months that would be typically necessary to significantly change the metabolic status.

ML methods have the potential to continuously evolve. In our scenario of MS prediction, a refinement of the model may be achieved by the incorporation of additional variables, such as diet patterns, different types of physical activity, fitness, and body composition, to name a few. However, the real impact of predicting MS by ML models on inducing a healthier lifestyle in the population is doubtful and should be investigated in future works. Additionally, the implementation of such tools in different settings should be dealt with caution, requiring external validation of our findings. Our results may not be generalizable to other populations. Although the algorithms were developed with a representative sample of the population in which they would be applied, it is expected a decline in prediction performance in novel circumstances because the correlation between predictive and causal factors may differ and the measurement of predictors and outcome may not be consistent [47].

There are limitations of this study. First, the observational design allowed us to investigate associations between potential predictors and the outcome and to emulate the effect of the interventions on the outcome. Randomization design would be more appropriate to ascertain a possible mechanistic role of the interventions on developing MS. Second, because only visit pairs were used to develop the algorithms, historical information generated by multiple and successive checkup visits was not considered. Future studies investigating algorithms that accommodate time series could clarify whether historical information would improve the prediction of future MS. Third, the large range of time between checkup visits was not optimal to assess the effect of lifestyle interventions. Fourth, we used single imputation to deal with missing data because it is simple to be implemented in the pre-processing pipeline. However, other techniques such as inverse probability weighting or multiple imputation could be used to avoid downplaying the variability between individuals. Finally, unmeasured confounding (e.g., diet effects) cannot be ruled out in the predictions.

In conclusion, we were able to develop and test models that presented good discrimination to predict future MS in adults attending a periodic health examination program using ML algorithms based on sociodemographic attributes, clinical, laboratory, and lifestyle characteristics. We also showed that these ML models could be used to estimate the effect of lifestyle modifications in the predicted probability of future MS. Applying these algorithms in clinical practice could be useful to identify individuals at higher risk that would benefit from early preventive measures and to engage such individuals in their own health management, by knowing their personalized risk and the simulated effect of lifestyle interventions. As well, these models might help the healthcare provider refine and personalize the care of patients. However, further studies to externally validate the algorithms should be performed before implementing the predictive models in different clinical settings.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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